Lesson 5.3 Guided Notes

Session 1 · Trapezoids by Decomposing

I can find the area of a trapezoid by decomposing it into a rectangle and triangles. · 6.G.A.1

Word / Spanish	Meaning — complete the blank	Example
trapezoid / trapecio	A _____ with one pair of parallel sides (the bases).	bases 6 cm and 14 cm
height / altura	The _____ distance between the two bases.	h, at a square corner
decompose / descomponer	To _____ a shape into smaller shapes you know.	rectangle + 2 triangles

Key idea

Cut a trapezoid into a _____ and one or two triangles.

Find the area of each piece, then _____ the areas.

Together, the triangle bases equal the _____ of the two bases.

Worked example

An isosceles trapezoid has bases 6 cm and 14 cm and a height of 5 cm. Find its area.

Rectangle: 6 × 5 = _____ cm².

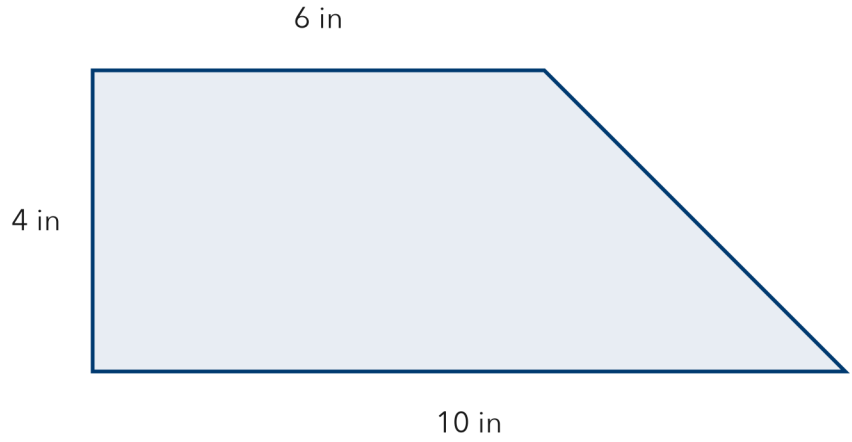
Leftover: 14 − 6 = 8, so each triangle base is _____ cm.

Both triangles: 2 × (½ × 4 × 5) = _____ cm².

Answer: Area = 30 + 20 = _____ cm²

You try

1. This trapezoid has one square corner. Decompose it and find its area.



_____

2. Each trapezoid has two matching end triangles. Find the base of one triangle.
a. bases 8 m and 14 m _____
b. bases 12 cm and 20 cm _____

_____

Sum it up: To find a trapezoid's area by decomposing, I ___ and then ___.

_____

Session 2 · The Trapezoid Area Formula

I can find the area of a trapezoid using the trapezoid area formula. · 6.G.A.1

Word / Spanish	Meaning — complete the blank	Example
base / base	One of the two _____ sides of a trapezoid.	b1 and b2
height / altura	The _____ distance between the two bases.	h
formula / fórmula	A _____ written with letters and numbers.	A = ½(b1 + b2)h
right trapezoid / trapecio rectángulo	A trapezoid with two _____ angles.	its straight side is the height

Key idea

Area of a trapezoid: A = ½ × (_____) × h.

_____ the bases first, then multiply by the height, then take half.

Two copies of a trapezoid make a _____, so the trapezoid is half of it.

Worked example

A trapezoid has bases 6 m and 12 m and a height of 4 m. Find its area.

Add the bases: 6 + 12 = _____.

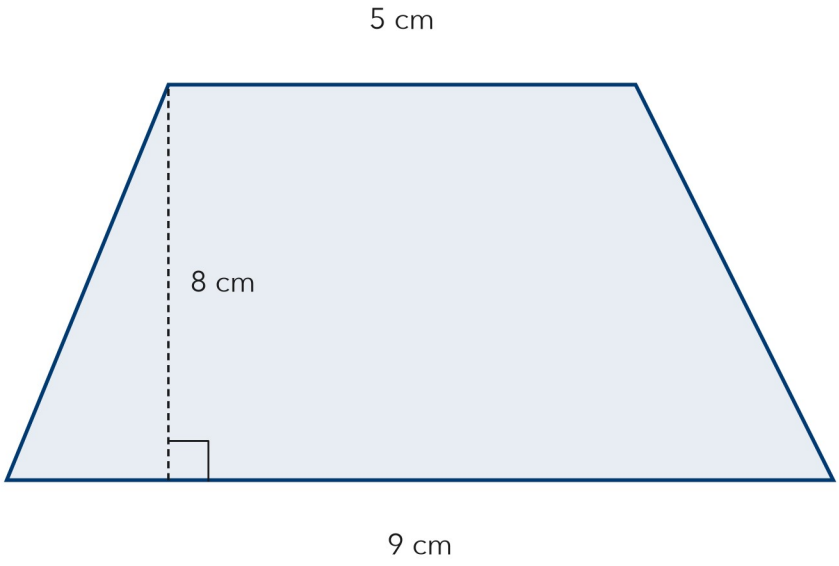
Multiply by the height: 18 × 4 = _____.

Take half: 72 ÷ 2 = _____.

Answer: A = _____ m²

You try

1. Find the area of the trapezoid.



_____

2. Find each area.
a. 3 m, 7 m; h = 4 m _____ m²
b. 10 in, 12 in; h = 5 in _____ in²

_____

Sum it up: To use the trapezoid formula, I ___ first because ___.

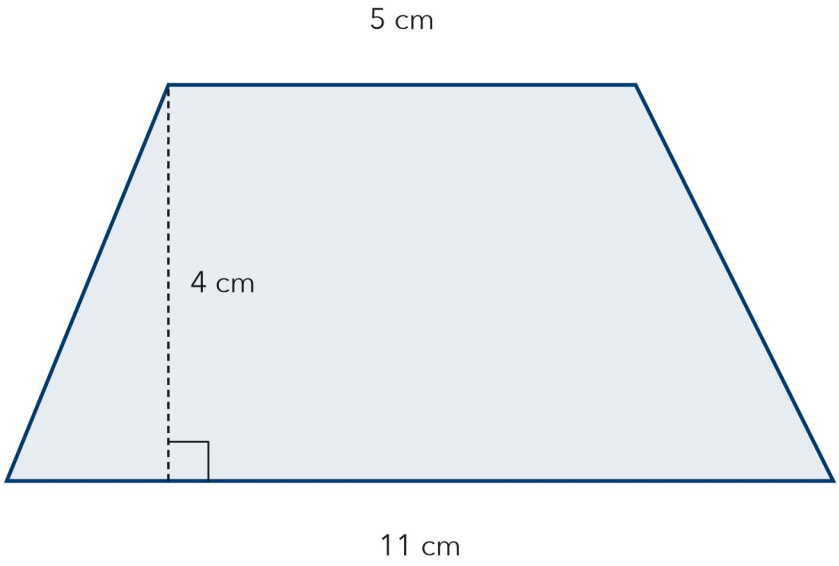
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Session 1 · Model, reason, and explain

Trapezoids by Decomposing · 6.G.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A trapezoid has parallel bases 5 cm and 11 cm and perpendicular height 4 cm. Draw two decompositions. Must the two leftover triangles have equal bases?



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: assumes all trapezoids are isosceles.. Explain what changes and what stays true.

_____

_____

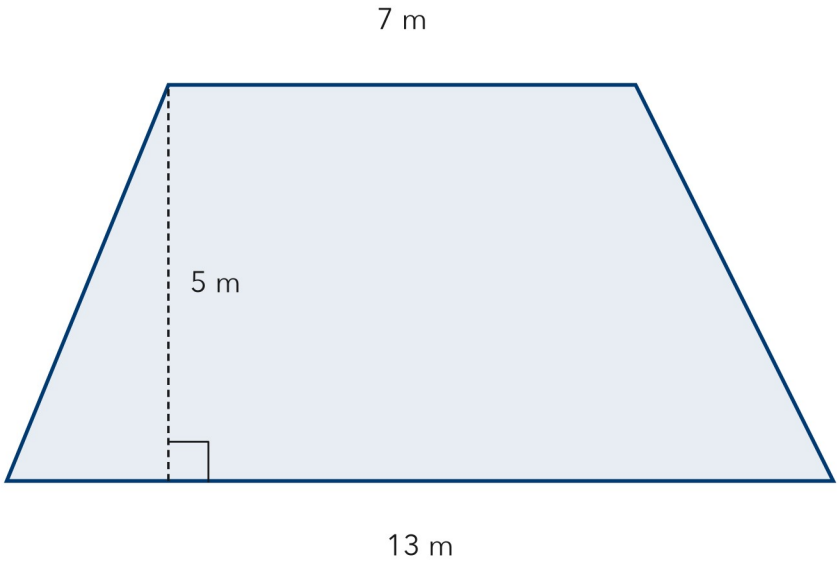
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

The Trapezoid Area Formula · 6.G.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Combine two congruent trapezoids with bases 7 m and 13 m and height 5 m into a parallelogram. Derive the area of one.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: adds the slanted sides instead of parallel bases.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?